

Akash Kokare

AI Engineer | Applied AI | Agentic Systems

📍 Mumbai, India 📞 +91-7400116992 ✉️ akashkokare2017@gmail.com

🌐 [linkedin.com/in/akash-kokare13bz](https://www.linkedin.com/in/akash-kokare13bz) 🐙 github.com/akashkokare2910 🌐 akashkokare2910.github.io

Summary

AI Engineer with 4+ years of experience building production AI systems and full-stack products. At **Birla AI Labs (Aditya Birla Group)**, develops the forecasting platform, multi-tenant public API, and agent tooling behind oab-forecasting.com, serving seven time-series foundation models through FastAPI and exposing forecasting workflows to Claude and ChatGPT through Model Context Protocol (MCP). Experienced in LLM and agent integration, AI-agent reliability/evaluation, ML inference infrastructure, data pipelines, and end-to-end delivery using Next.js, FastAPI, and AWS.

Technical Skills

AI / ML: LLM Integration, Agentic Workflows, AI Agent Evaluation, Model Context Protocol (MCP), RAG, Prompt Engineering, LangChain, Time-Series Foundation Models (Chronos, TimesFM, Moirai), Amazon Bedrock, SageMaker, Model Deployment, TensorFlow

Backend / Infra: FastAPI, Python, Redis, Async Job Queues, Multi-Tenant APIs, REST, WebSockets, Microservices, Node.js, JWT / OAuth

Frontend: Next.js (App Router), React, TypeScript, Redux / RTK Query, TailwindCSS

Cloud / DevOps: AWS (EC2, S3, Lambda, RDS, Bedrock), Docker, CI/CD (GitHub Actions, SSM), PyPI Trusted Publishing, Git, Serverless

Data: PostgreSQL, Supabase, DynamoDB, S3 / Parquet, Elasticsearch, Vector Databases, MongoDB, MySQL

Languages: Python, TypeScript, JavaScript, Java, C/C++

Experience

AI Engineer

Feb 2025 – Present

Birla AI Labs - Office of Ananya Birla, Aditya Birla Group

Mumbai, India

- Built the forecasting platform behind **oab-forecasting.com** (Next.js 14 App Router, FastAPI, Supabase, AWS): dataset upload for CSV/Excel/Parquet/JSON, an ingestion pipeline that normalises, validates, imputes gaps, and infers schema, and server-side inference across **seven zero-shot time-series foundation models** (Chronos-2, TimesFM-2, Moirai-2, TabPFN-TS, and an in-house fine-tuned **Bharat-TSFM** for Indian-market data).
- Designed a **multi-tenant public forecasting API** in FastAPI with API-key and OAuth auth, per-org quotas, rate limiting, and a **Redis-backed async job queue** with idempotency keys that survives restarts and is shared across containers; exposed it to **Claude and ChatGPT via a Model Context Protocol (MCP) server**.
- Built the **Forecast Agent**: a GPT-4o chat panel answering natural-language questions over forecast results, plus PDF export, connecting model output directly to product UX.
- Delivered UltraTech Cement's demand-forecasting system: **hierarchical forecasting** across SKU and region, calendar/holiday/weather feature engineering, an inventory optimiser, and forecast snapshotting with run-to-run diffing; integrated with **Anaplan** for enterprise planning.
- Built the **time-series data engineering layer**: ingestion, validation, feature generation, dataset versioning, and S3-backed transactional sync, feeding foundation-model training and inference.
- Owned Dockerised **CI/CD to AWS EC2** via GitHub Actions and SSM across test/dev/prod, with strict typing (mypy) and linting (ruff) enforced in the pipeline.

Software Engineer

Jul 2022 – Dec 2024

ResearchWire Knowledge Solutions Pvt. Ltd.

Navi Mumbai, India

- Built **ResearchWire Analytics Platform** (React, Symphony, MySQL) with Elasticsearch integration for querying millions of patent documents; improved search efficiency and result accuracy by **30%**.
- Optimised XML-to-JSON conversion pipeline using XSLT and Java (Spring Boot); reduced processing time by **50%**, enabling faster Elasticsearch indexing and query execution.
- Re-architected conversion pipeline in Node.js, achieving **10x faster throughput**; improved horizontal scalability of the Elasticsearch ingestion system.
- Built MERN-based IDS docketing platform with RTK Query, Spring Boot PDF generation, and real-time data retrieval from USPTO and Espace patent APIs.

Machine Learning Intern

Jun 2021 – Aug 2021

Regex Software Services

Remote

- Fine-tuned **CNN garbage classification model** using InceptionV3 transfer learning; achieved **90% validation accuracy** through data preprocessing, feature extraction, and hyperparameter tuning.

Projects

EvalForge - AI Agent Reliability & Evaluation Platform | *Python, FastAPI, PostgreSQL, Next.js, OpenAI, Supabase, GitHub Actions* [Live] | [GitHub]

- Built and deployed a stateful reliability-testing platform for tool-using AI agents, evaluating tool selection, arguments, execution trajectories, final environment state, failure categories, and baseline-vs-candidate regressions instead of scoring only final responses.
- Engineered a multi-tenant FastAPI/PostgreSQL backend with workspace authorization and IDOR protection, scoped API tokens, secret-safe trace ingestion, encrypted workspace BYOK OpenAI credentials, and a Python SDK/CLI plus GitHub Actions quality gates for external agents and CI.
- Ran a manually audited 20-case real-model pilot across **gpt-4o-mini** and **gpt-4o**, surfacing unsafe/incorrect state mutations invisible to final-answer scoring; backed by **847 automated tests** and a live Next.js/Vercel + FastAPI/Render + Supabase deployment.

MCPLint - Behavioral Reliability Testing for MCP Agent Tools | *Python, Pydantic, Typer, Anthropic API, GitHub Actions* [GitHub] | [PyPI: mcplint-cli]

- Designed and published an open-source CLI (`pip install mcplint-cli`, live on PyPI, v0.2.0) that detects why LLM agents pick the wrong Model Context Protocol (MCP) tool: 15 deterministic lint rules, a cross-tool ambiguity engine, and an explainable 0–100 score, with SARIF/HTML/JSON reporters for CI gating.
- Built **ContractLab**: a behavioral-contract spec (`mcplint.contract.yaml`), adversarial benchmark generation from contract rules, mutation testing that verifies tool descriptions actually carry the distinctions they claim, and confusion analysis cross-referencing static ambiguity predictions against real observed model tool-selection errors; 220+ automated tests, fully deterministic scoring, no LLM judge.
- Owned production release engineering end to end: PyPI Trusted Publishing (OIDC, zero stored tokens), GitHub Actions CI/CD with manual approval gates, semantic versioning, and signed package attestations on every release.

RAG-Powered Document Assistant | *Next.js, FastAPI, LangChain, OpenAI, Supabase pgvector*

- Built end-to-end RAG pipeline: document ingestion, chunking, embedding, vector storage (pgvector), retrieval, and LLM synthesis; deployed on AWS with FastAPI backend and Next.js chat UI.
- Implemented semantic similarity search with MMR re-ranking, cutting irrelevant retrieval by 40% vs naive top-k.

Education

Atharva College of Engineering, Mumbai
Bachelor of Engineering - Information Technology

Jul 2018 – Jul 2022

Certifications

AWS Certified Machine Learning Engineer - Associate (MLA-C01) (*in progress*)

AWS Solutions Architect Associate Course (*Udemy*)

AWS Certified Cloud Practitioner (*in progress*)

Machine Learning A-Z: AI, Python and R + ChatGPT

ReactJS - Namaste Dev